# BATCH7_RTA_GEOM

# PREREG: BATCH7_RTA_GEOM (Map-C25 to Map-C27): $K^+$ homology over F_p, p | 140 vs p ∤ 140

Written 2026-10-24 at about 7:15 PM CDT, before any $K^+$ computation. Owner: Arvin Hampton. New files only. CORE_FREEZE. All maps are **Category B**; vacuous passes are flagged. order21 and the paper are not touched. C15–C24 are not reopened. No Π or grading is invented.

## Modular data (generated by script, not typed by hand)

scripts/modular_table.py (sha256 b994249512f4fc869034e62375cfc37920af72bacf4da8309a80231c53b3bc7f) → data/modular_table.json (sha256 9201e860b6184c0bc82cfd0adf2e6ac7cb347fb340cea758decd4cd386017309). Contents, copied from the JSON: - p | 140: {2, 5, 7}. - Control p ∤ 140: {3, 11, 13, 17, 19, 23, 29}. - Inert ($\chi_{140}(p) = -1$): {3, 11}. Split: {13, 17, 19, 23, 29}.

## Complex $K^+$ (written; not invented)

build_complex(O, loads, “A2_enrich”) from /tmp/repo539/docs/S2-11DM2ET-X-Axiomatic-Book/scripts/architecture_A4plus_probe.py, imported read-only. It is the path plus all same-tower edges plus the 2-fill of all triangles in that graph. Gate (else VOID): V = 539, E = 594, F = 56 (written). Exact method: ranks of $\partial_1$ (E × V) and $\partial_2$ (F × E) with signed (oriented) incidences, over Q (fmpz) and over F_p (nmod). β_k(F_p) from the ranks. Integer torsion at p is detected by rank_p < rank_Q.

## Map-C25 (primary; the 140 split)

Prediction: there exists $p \in \{2, 5, 7\}$ with $(\beta_0, \beta_1, \beta_2)(F_p) \neq (\beta_0, \beta_1, \beta_2)(F_3)$, while all control primes give the same Betti vector as F_3. PASS iff both hold.

## Map-C26 (torsion support)

Prediction: H_*($K^+$; Z) has non-trivial torsion, and every torsion prime lies in {2, 5, 7} (among all tested primes). PASS iff torsion ≠ 0 and its tested support ⊆ {2, 5, 7}.

## Map-C27 (form spectral sequence over F_p). OPEN by construction

Residual_Form_Spectral_Sequences.md defines $E_1/E_2$ only over Q, with generators listed symbolically. No F_p generator matrices are written. Prediction (to be tested once matrices are written): over F_p with p | 140 the SS fails to collapse at $E_2$; for p ∤ 140 it collapses. Blocker: written F_p differential matrices for the thin form complex.

## Still OPEN: C6, C10, C11, E1–E3.

# RESULTS: BATCH7_RTA_GEOM

PREREG.md sha256 f817ce4d2c613813e01198c75ad4b65f63581deeaf77efe048fb01c9c1509a33, hashed after the modular-table script and before any $K^+$ computation. Scripts: scripts/modular_table.py, scripts/b7.py (hashes in scripts/SHA256SUMS). Data: data/modular_table.json, data/b7.json. Logs: logs/. Exact ranks (python-flint, fmpz and nmod), with $\partial_2\partial_1 = 0$ asserted. Under 1 s.

Gate: (V, E, F) = (539, 594, 56). OK. β(Q) = (1, 0, 0). β(F_p) = (1, 0, 0) for every p ∈ {2, 5, 7, 3, 11, 13, 17, 19, 23, 29}. No rank drop at any tested prime, so there is no torsion at the tested primes.

| Map | Prediction | Measured | Verdict | Vacuous? |
|---|---|---|---|---|
| C25 | β(F_p) differs for some p \| 140 vs the controls | identical at all primes | **FAIL** | — |
| C26 | torsion ≠ 0, supported in {2, 5, 7} | no torsion at tested primes | **FAIL** | — |
| C27 | form SS over F_p fails to collapse iff p \| 140 | no written F_p matrices | **OPEN** | — |
| C6, C10, C11, E1–E3 | — | blocked | OPEN | — |

**Bridge: NO.**

Post-hoc (Category A remark, no verdict change): the written facts $\beta = (1,0,0)$ over $F_2$ with $\chi = 1$, plus no rank drop at the 10 tested primes, mean $K^+$ is Z-acyclic at those primes. Full Z-acyclicity would need either a Smith normal form or the remaining primes dividing the elementary divisors; a contractibility argument (already claimed in the written lock) would cover it. $K^+$ homology is blind to every prime, so no modular split can be read from it. # NOTES: BATCH7_RTA_GEOM ## Sources - $K^+$ complex: /tmp/repo539/docs/S2-11DM2ET-X-Axiomatic-Book/scripts/architecture_A4plus_probe.py (build_complex mode “A2_enrich”, atoms, core_q0; imported read-only). Docs: Architecture_A4plus_Continuous_BSpin.md, Architecture_A4_BSpin_Lift.md, architecture_A4plus_results.json (same folder). Counts $V = 539$, $E = 594$, $F = 56$ and $\beta = (1,0,0)$ over $F_2$: grok-chats/13-t3-vs-hqcc.md line 45. - Form SS: /tmp/repo539/docs/S2-11DM2ET-X-Axiomatic-Book/Residual_Form_Spectral_Sequences.md ($E_1/E_2$ over Q; collapse at $E_2$, lines 205–207) and Thin_Complex_Formality.md §1.

## Process fix (as required)

All modular numbers in the PREREG come from scripts/modular_table.py → data/modular_table.json, both hashed in the PREREG. Nothing was typed by hand.

## Remarks

- The $K^+$ boundary orientation is the standard simplicial one (sorted vertices). Over $F_2$ it agrees with the probe’s GF(2) incidence, and it reproduces the written $\beta$ over $F_2$.
- Pattern: $K^+$, $P^+$, the permanent-class survival test and the fibre counts are all either contractible/acyclic or carrier-forced. The residual topology carries no prime-dependent or character-dependent signal for the 35-component law vector data to hit.

## Sharpest next step

- The residual-internal levers are now exhausted in an honest sense: the written residual complexes are acyclic over every tested prime, and the cheap invariants are carrier-forced.
- The only remaining non-vacuous route is the **Resonant side**, which needs a written object: a 539-step $\Pi$ word, a $\pm 1$ grading on O_res, or a path predicate (C6/C10/C11/E1–E3).
- Alternative: written F_p differentials for the thin form complex (unblocks C27).
- Recommend pausing new residual-side maps until one of these is written by Arvin.
- order21 and the paper were not touched. CORE_FREEZE; new files only; no push, no sends.